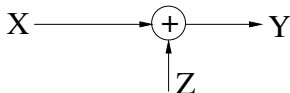


Continuous valued channels

AWGN Channel



$$Y = X + Z \quad Z \sim \mathcal{N}(0, \sigma^2)$$

- Need to study information in continuous valued random variables.
- How much information does a continuous random variable contain?

Differential entropy

where S denotes the support of f .

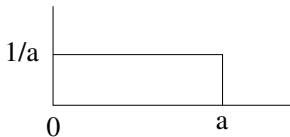
- We assume that the density of X exists, and the above integral exists.
- This does not measure the information in X .

Joint differential entropy

$$h(X_1, \dots, X_n) = - \int f(x^n) \log f(x^n) dx^n$$

One can easily see that $h(X|Y) = h(X, Y) - h(Y)$.

Example: uniform distribution



$$h(X) = - \int_0^a \frac{1}{a} \log \frac{1}{a} dx = \log a$$

Normal distribution: $X \sim \mathcal{N}(0, \sigma^2)$

$$\begin{aligned}
 h(X) &= - \int_{-\infty}^{\infty} \phi(x) \ln \phi(x) dx \quad \text{nats} \\
 &= - \int_{-\infty}^{\infty} \phi(x) \left[-\frac{x^2}{2\sigma^2} - \ln \sqrt{2\pi\sigma^2} \right] dx \\
 &= \frac{E[X^2]}{2\sigma^2} + \frac{1}{2} \ln 2\pi\sigma^2 \\
 &= \frac{1}{2} + \frac{1}{2} \ln 2\pi\sigma^2 \\
 &= \frac{1}{2} \ln e + \frac{1}{2} \ln 2\pi\sigma^2 \\
 &= \frac{1}{2} \ln 2\pi e\sigma^2 \quad \text{nats} \\
 &= \frac{1}{2} \log 2\pi e\sigma^2 \quad \text{bits}
 \end{aligned}$$

Properties

$$h(X + c) = h(X)$$

-follows from definition

$$h(aX) = h(X) + \log |a|$$

Proof: Let $Y = aX$. Then $f_Y(y) = (1/|a|)f_X(y/a)$.

$$\begin{aligned} h(aX) &= - \int f_Y(y) \log f_Y(y) dy \\ &= - \int \frac{1}{|a|} f_X\left(\frac{y}{a}\right) \log \left(\frac{1}{|a|} f_X\left(\frac{y}{a}\right)\right) dy \\ &= - \int f_X(x) \log f_X(x) dx + \log |a| \\ &= h(X) + \log |a| \end{aligned}$$

Properties

More generally, for an $n \times n$ matrix A ,

$$h(A\mathbf{X}) = h(\mathbf{X}) + \log |\det(A)|$$

A hand-drawn graph of a probability density function $f(x)$ versus x . The curve is bell-shaped and symmetric. The area under the curve is divided into several vertical strips. The width of one strip is labeled Δ . The height of the strip is labeled p_i . The x-axis is labeled x .

Information in a continuous random variable

Consider the quantized version X^Δ of X , where Δ is the step size of the uniform quantization.

$$X^\Delta = x_i \quad \text{if } i\Delta \leq X \leq (i+1)\Delta$$

$$f(x_i)\Delta = \int_{i\Delta}^{(i+1)\Delta} f(x)dx$$

$$p_i := \Pr(X^\Delta = x_i) = \int_{i\Delta}^{(i+1)\Delta} f(x)dx = f(x_i)\Delta$$

$$\begin{aligned} H(X^\Delta) &= -\sum p_i \log p_i \\ &= -\sum f(x_i)\Delta \log(f(x_i)\Delta) \\ &= -\sum \Delta f(x_i) \log f(x_i) - \sum f(x_i)\Delta \log \Delta \end{aligned}$$

Information in a continuous random variable

$$\begin{aligned}
 H(X^\Delta) &= - \sum \Delta f(x_i) \log f(x_i) - \sum f(x_i) \Delta \log \Delta \\
 &= - \sum \Delta f(x_i) \log f(x_i) - \log \Delta
 \end{aligned}$$

Thus

$$\begin{aligned}
 H(X^\Delta) + \log \Delta &= - \sum \Delta f(x_i) \log f(x_i) \\
 &\rightarrow h(X) \quad \text{as } \Delta \rightarrow 0
 \end{aligned}$$

AEP for continuous random variables

$X_1, X_2, \dots, X_n$ i.i.d. $\sim f(x)$

$-\frac{1}{n} \log f(X_1, \dots, X_n) \rightarrow E[-\log f(X)] = h(X)$ in probability

Typical set

$$A_\epsilon^{(n)} := \{(x_1, \dots, x_n) \in \mathcal{S}^n : \left| -\frac{1}{n} \log f(x_1, \dots, x_n) - h(X) \right| \leq \epsilon\}$$

where $f(x_1, \dots, x_n) = \prod_{i=1}^n f(x_i)$.

Typical set and balls: Gaussian example

For a Gaussian random variable X ,

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}}$$

$$h(X) = \frac{1}{2} \log(2\pi e\sigma^2) \text{ bits}$$

$$\begin{aligned} -\frac{1}{n} \log f(x^n) &= -\frac{1}{n} \log \left(\prod_{i=1}^n f(x_i) \right) \\ &= -\frac{1}{n} \log \left(\prod_{i=1}^n (2\pi\sigma^2)^{-1/2} e^{-\frac{x_i^2}{2\sigma^2}} \right) \\ &= -\frac{1}{n} \log \left((2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2} \sum x_i^2} \right) \\ &= \frac{1}{2} \log(2\pi\sigma^2) + \frac{\log e}{2n\sigma^2} \sum x_i^2 \end{aligned}$$

Typical set

$\mathbf{x}$ is typical

$$\Rightarrow -\frac{1}{n} \log f(x^n) \approx \frac{1}{2} \log(2\pi e\sigma^2)$$

$$\Rightarrow \frac{1}{2} \log(2\pi\sigma^2) + \frac{\log e}{2n\sigma^2} \sum x_i^2 \approx \frac{1}{2} \log(2\pi e\sigma^2)$$

$$\Rightarrow \frac{\log e}{2n\sigma^2} \sum x_i^2 \approx \frac{1}{2} \log(e)$$

$$\Rightarrow \sum x_i^2 \approx n\sigma^2$$

This means that $\mathbf{x}$ is on the surface of a sphere of radius $\sqrt{n}\sigma$.

Typical set

For any $A \subset \mathbb{R}^n$, its volume is defined as

$$\text{Vol}(A) := \int_A dx_1 \cdots dx_n$$

Theorem

- ① $\Pr \left(A_\epsilon^{(n)} \right) > 1 - \epsilon$ for n sufficiently large.
- ② $\text{Vol} \left(A_\epsilon^{(n)} \right) \leq 2^{n(h(X)+\epsilon)} \quad \forall n.$
- ③ $\text{Vol} \left(A_\epsilon^{(n)} \right) \geq (1 - \epsilon)2^{n(h(X)-\epsilon)} \quad \text{for large enough } n.$

Proof similar to the discrete case, and can be found in Cover and Thomas.

- If $A_\epsilon^{(n)}$ is thought as an n -D cube, then $2^{h(X)}$ is the length of each side.

Conditional differential entropy

Conditional differential entropy

$$h(X|Y) = - \int f(x, y) \log f(x|y) dx dy$$

One can easily see that $h(X|Y) = h(X, Y) - h(Y)$.

Multivariate normal distribution

$K = E[(\mathbf{X} - \mu)(\mathbf{X} - \mu)^T]$: covariance matrix

$$f(x_1, x_2, \dots, x_n) = \frac{1}{(\sqrt{2\pi})^n |K|^{\frac{1}{2}}} \exp \left(-\frac{1}{2} (x - \mu)^T K^{-1} (x - \mu) \right)$$

$$h(\mathbf{X}) = \frac{1}{2} \log(2\pi e)^n |K| \text{ bits}$$

See derivation in the book by Cover and Thomas.

KL divergence and mutual information

Let f, g be two densities.

$$D(f||g) := \int f \log \frac{f}{g}$$

where $0 \log \frac{0}{0} := 0$. The integral exists only if $\text{supp}(f) \subseteq \text{supp}(g)$.

$$I(X; Y) := D(f(x, y) || f(x)f(y))$$

Mutual information

If X^Δ, Y^Δ denote the random variables uniformly quantized with step size Δ , then

$$\begin{aligned}
 I(X^\Delta; Y^\Delta) &= H(X^\Delta) - H(X^\Delta | Y^\Delta) \\
 &\simeq h(X) - \log \Delta - (h(X|Y) - \log \Delta) \\
 &= I(X; Y)
 \end{aligned}$$

Mutual information

More general definition

For any two random variables X, Y (with or without densities), let $\mathcal{P}, \mathcal{Q}$ be partitions of $\mathcal{X}, \mathcal{Y}$ respectively. Let $X_{\mathcal{P}}, Y_{\mathcal{Q}}$ denote the quantized versions of X, Y respectively. Then

$$I(X; Y) := \sup_{\mathcal{P}, \mathcal{Q}} I(X_{\mathcal{P}}; Y_{\mathcal{Q}})$$

For X, Y with densities, this definition gives the earlier definition.

Mutual information

Theorem

$$D(f||g) \geq 0$$

with “=” iff $f = g$ almost everywhere.

Corollaries

$I(X; Y) \geq 0$; “=” iff X, Y are ind.

$h(X|Y) \leq h(X)$; “=” iff X, Y are ind.

$$h(X_1, X_2, \dots, X_n) = \sum_{i=1}^n h(X_i | X_1, \dots, X_{i-1})$$

$$h(X_1, X_2, \dots, X_n) \leq \sum_{i=1}^n h(X_i)$$

Gaussian has the maximum diff entropy

Let $\mathbf{X} \in \mathbb{R}^n$ be zero mean and covariance $K = E\mathbf{X}\mathbf{X}^t$ (i.e. $K_{ij} = EX_iX_j$). Then $h(\mathbf{X}) \leq \frac{1}{2} \log(2\pi e)^n |K|$, with equality iff $\mathbf{X} \sim \mathcal{N}(0, K)$.

Proof: Let ϕ_K denote the density of $\mathcal{N}(0, K)$. Let X have density $g(\mathbf{x})$ with $\mu = 0$ and covariance K .

Note: $\log \phi_K(\mathbf{x})$ is a quadratic form, and $\int x_i x_j \phi_K(\mathbf{x}) d\mathbf{x} = K_{ij}$.

$$\begin{aligned} 0 &\leq D(g||\phi_K) \\ &= \int g \log(g/\phi_K) \\ &= -h(g) - \int g \log \phi_K \end{aligned}$$

Gaussian has the maximum diff entropy

$$\begin{aligned}
 0 &\leq -h(g) - \int g \log \phi_K \\
 &= -h(g) - \int \phi_K \log \phi_K \\
 &= -h(g) + h(\phi_K) \\
 \Rightarrow h(g) &\leq h(\phi_K)
 \end{aligned}$$

Here $\int g \log \phi_K = \int \phi_K \log \phi_K$ follows from the fact that the quadratic form $\log \phi_K(\mathbf{X})$ has the same expectation under ϕ_K and g .

Determinant inequalities from differential entropy of Gaussian r.v.

Hadamard inequality

If K is a covariance matrix (positive semidefinite), then

$$|K| \leq \prod_{i=1}^n K_{ii}$$

Proof: Consider $\mathbf{X} \sim \mathcal{N}(0, K)$ and use $h(\mathbf{X}) \leq \sum h(X_i)$.

Convexity / concavity properties

Convexity of KL divergence

If f, g are density functions, then $D(f||g)$ is a convex function of the pair (f, g)

The proof follows in the same way as in the discrete case; using Log-sum inequality inside the integral.
See Th. 2.7.2 in Cover and Thomas.

Convexity / concavity properties

Concavity of differential entropy

If f denotes a density function, then $h(f)$ is a concave function of f .

This does not follow from the convexity of KL divergence as in the discrete case (see the proof of Th. 2.7.3 in Cover and Thomas). The difficulty lies in taking a suitable uniform distribution. But the alternative proof in Cover and Thomas works. The proof is given below.

Convexity / concavity properties

Proof: First, note that conditioning reduces differential entropy.

$$\begin{aligned} h(X) - h(X|Y) &= I(X; Y) \geq 0 \\ \Rightarrow h(X) &\geq h(X|Y) \end{aligned}$$

Suppose X_1 and X_2 have the density functions respectively $f_1(x)$ and $f_2(x)$. Define a new random variable θ which takes the values 1, 2 with probabilities respectively λ and $1 - \lambda$. Now define $Z = X_\theta$.

Then

$$\begin{aligned} h(Z) &\geq h(Z|\theta) \\ \Rightarrow h(\lambda f_1 + (1 - \lambda)f_2) &\geq \lambda h(f_1) + (1 - \lambda)h(f_2) \end{aligned}$$

This proves the result.

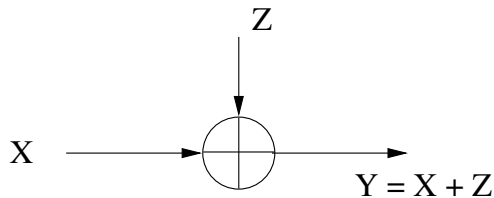
Convexity / concavity properties

Concavity/Convexity of mutual information

$I(X; Y)$ is a concave function of $f(x)$ for fixed $f(y|x)$, and a convex function of $f(y|x)$ given $f(x)$.

The same proof as in the discrete case works; see the proof of Th. 2.7.4 in Cover and Thomas.

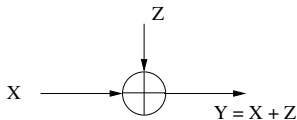
AWGN Channel



$$Z \sim \mathcal{N}(0, \sigma^2)$$

What is the capacity?

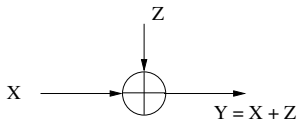
AWGN Channel



Input power constraint:

$$\frac{1}{n} \sum_{i=1}^n x_i^2 \leq P$$

AWGN Channel



Input power constraint:

$$\frac{1}{n} \sum_{i=1}^n x_i^2 \leq P$$

Sphere packing: A rough calculation

Noise vector length $\simeq \sqrt{n}\sigma$.

$\Rightarrow$ Decoding sphere radius $r \simeq \sqrt{n}\sigma$.

Received vector length $R \simeq \sqrt{n(P + \sigma^2)}$.

Number of spheres packed

$$\begin{aligned} \frac{V_n(R)}{V_n(r)} &= \frac{R^n}{r^n} \\ &= \left(\frac{P + \sigma^2}{\sigma^2} \right)^{n/2} \end{aligned}$$

$$\text{Rate} = \frac{1}{2} \log \left(1 + \frac{P}{\sigma^2} \right)$$

Sphere packing

What can be achieved by sphere packing:

$$\frac{1}{2} \log \left(1 + \frac{P}{\sigma^2} \right) - \frac{1}{2}$$

For a more precise discussion, see my handwritten note.

Code

Code: An (M, n) code with power constraint P consists of

- ① Index set $[1 : M] = \{1, 2, \dots, M\}$
- ② Encoding function $x : \{1, 2, \dots, M\} \rightarrow \mathcal{X}^n$, giving codewords $\mathbf{x}(1), \mathbf{x}(2), \dots, \mathbf{x}(M)$, satisfying power constraint

$$\sum_{i=1}^n x_i^2(w) \leq nP, \quad w \in [1 : M]$$

- ③ Decoding function

$$g : \mathcal{Y}^n \rightarrow [1 : M]$$

Rate of a code, P_e , achievable rate, capacity: defined as in the discrete case

Coding theorem:

The capacity of the AWGN channel with power constraint P is given by

$$C = \max_{EX^2 \leq P} I(X; Y) = \frac{1}{2} \log \left(1 + \frac{P}{\sigma^2} \right)$$

Maximum is achieved by $X \sim \mathcal{N}(0, P)$.

Note: $\text{Var}(Y) = \text{Var}(X) + \text{Var}(Z) \leq P + \sigma^2$.

$$\begin{aligned} I(X; Y) &= h(Y) - h(Y|X) = h(Y) - h(Y - X|X) \\ &= h(Y) - h(Z|X) = h(Y) - h(Z) \\ &\leq \frac{1}{2} \log_2(2\pi e(\sigma^2 + P)) - \frac{1}{2} \log_2(2\pi e\sigma^2) \\ &= \frac{1}{2} \log_2 \left(1 + \frac{P}{\sigma^2} \right) \end{aligned}$$

Coding theorem:

The capacity of the AWGN channel with power constraint P is given by

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Achievability

- 1 **Code construction:** Take any $R < C$, and construct a code by picking each component i.i.d. $\sim \mathcal{N}(0, P - \epsilon)$. Since $\frac{1}{n} \sum X_i^2 \rightarrow P - \epsilon$, there will be only a small fraction of codewords violating the power constraint P .
- 2 **Encoding:** For encoding w , we send the w -th codeword if it satisfies the power constraint. Otherwise we transmit the all-zeros vector.
- 3 **Decoding:** Upon receiving $\mathbf{Y}$, we do joint typicality decoding.

Probability of error:

We study the expected $P_e^{(n)}$ over the code randomness. This is same for all messages w , so we assume $w = 1$.

Let us define the events

$$\begin{aligned} E_0 &:= \left\{ \frac{1}{n} X_j(1) > P \right\} \\ E_i &:= \left\{ (\mathbf{X}(i), \mathbf{Y}) \in A_\epsilon^{(n)} \right\}, \quad 1 \leq i \leq M \end{aligned}$$

Then

$$\begin{aligned} Pr(Error|W = 1) &= Pr(E_0 \cup E_1^c \cup E_2 \cup \dots \cup E_M) \\ &\leq Pr(E_0) + Pr(E_1^c) + \sum_{i=2}^M Pr(E_i) \end{aligned}$$

Probability of error:

- 1 $Pr(E_0) \rightarrow 0$ by WLLN
- 2 $Pr(E_1^c) \rightarrow 0$ by AEP
- 3 For $i \neq 1$,

$$Pr(E_j) \leq 2^{-n(I(X;Y)-3\epsilon)}$$

since $\mathbf{X}(i)$, $\mathbf{Y}$ are independently generated.

Probability of error:

$$\begin{aligned} \Pr(\text{Error} | W = 1) &\leq \Pr(E_0) + \Pr(E_1^c) + \sum_{i=2}^M \Pr(E_i) \\ &\leq \epsilon + \epsilon + \sum_{i=2}^M 2^{-n(I(X;Y)-3\epsilon)} \\ &\leq 2\epsilon + 2^{nR} \cdot 2^{-n(I(X;Y)-3\epsilon)} \\ &< 2\epsilon + 2^{3\epsilon n} \cdot 2^{-n(I(X;Y)-R)} \end{aligned}$$

Probability of error:

For $X \sim \mathcal{N}(0, P - \epsilon)$,

$$I(X; Y) \geq C - f(\epsilon)$$

for f s.t. $f(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$.

So,

$$\begin{aligned} Pr(Error|W = 1) &\leq 2\epsilon + 2^{3\epsilon n} \cdot 2^{-n(C-R-f(\epsilon))} \\ &\leq 2\epsilon + 2^{3\epsilon n} \cdot 2^{-n(C-R-f(\epsilon))} \\ &= 2\epsilon + 2^{(f(\epsilon)+3\epsilon)n} \cdot 2^{-n(C-R)} \\ &< 3\epsilon \end{aligned}$$

for sufficiently large n , and for $R < C - f(\epsilon) - 3\epsilon$.

Coding theorem:

The capacity of the AWGN channel with power constraint P is given by

$$C = \max_{EX^2 \leq P} I(X; Y) = \frac{1}{2} \log \left(1 + \frac{P}{\sigma^2} \right)$$

Maximum is achieved by $X \sim \mathcal{N}(0, P)$.

Note: $Var(Y) = Var(X) + Var(Z) \leq P + \sigma^2$.

$$\begin{aligned} I(X; Y) &= h(Y) - h(Y|X) = h(Y) - h(Y - X|X) \\ &= h(Y) - h(Z|X) = h(Y) - h(Z) \\ &\leq \frac{1}{2} \log_2(2\pi e(\sigma^2 + P)) - \frac{1}{2} \log_2(2\pi e\sigma^2) \\ &= \frac{1}{2} \log_2 \left(1 + \frac{P}{\sigma^2} \right) \end{aligned}$$

Converse proof

Using this, it follows that

$$\begin{aligned} I(X; YZ) &= h(X) - h(X|Y, Z) \\ &= (h(X) - h(X|Y)) + (h(X|Y) - h(X|Y, Z)) \\ &= I(X; Y) + I(X; Z|Y) \end{aligned}$$

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$$I(W; \hat{W}) \leq I(\mathbf{X}; \mathbf{Y})$$

100

For a sequence of codes of rate $\geq R$ satisfying the power constraint, and with $P_e^{(n)} \rightarrow 0$, we want to show that $R \leq C$.

Fano's inequality gives

$$H(W|\hat{W}) \leq 1 + nRP_e^{(n)} = n\epsilon_n$$

for some $\epsilon_n \rightarrow 0$ as $n \rightarrow 0$.

Converse

$$\begin{aligned} nR &= H(W) \\ &= I(W; \hat{W}) + H(W | \hat{W}) \\ &\leq I(W; \hat{W}) + n\epsilon_n \\ &\leq I(\mathbf{X}; \mathbf{Y}) + n\epsilon_n \\ &= h(\mathbf{Y}) - h(\mathbf{Y} | \mathbf{X}) + n\epsilon_n \\ &= h(\mathbf{Y}) - h(\mathbf{Z}) + n\epsilon_n \\ &\leq \sum_{i=1}^n h(Y_i) - h(\mathbf{Z}) + n\epsilon_n \\ &= \sum_{i=1}^n h(Y_i) - \sum_{i=1}^n h(Z_i) + n\epsilon_n \end{aligned}$$

Converse

$$\begin{aligned} nR &= \sum_{i=1}^n h(Y_i) - \sum_{i=1}^n h(Z_i) + n\epsilon_n \\ &= \sum_{i=1}^n I(X_i; Y_i) + n\epsilon_n \end{aligned}$$

Define the average power transmitted in the i -th component as

$$P_i = EX_i^2 := 2^{-nR} \sum_w x_i^2(w).$$

Then

$$h(Y_i) \leq \frac{1}{2} \log 2\pi e(P_i + \sigma^2).$$

Converse

$$\begin{aligned}
 nR &= \sum_{i=1}^n (h(Y_i) - h(Z_i)) + n\epsilon_n \\
 &\leq \sum_{i=1}^n \left(\frac{1}{2} \log 2\pi e(P_i + \sigma^2) - \frac{1}{2} \log 2\pi e\sigma^2 \right) + n\epsilon_n \\
 &\leq \sum_{i=1}^n \frac{1}{2} \log \left(1 + \frac{P_i}{\sigma^2} \right) + n\epsilon_n
 \end{aligned}$$

Since all the codewords satisfy the power constraint,

$$\frac{1}{n} \sum_i P_i \leq P$$

Converse

Since $(1/2) \log(1 + x)$ is a concave function of x , we have

$$\begin{aligned}
 nR &\leq \sum_{i=1}^n \frac{1}{2} \log \left(1 + \frac{P_i}{\sigma^2} \right) + n\epsilon_n \\
 &\leq \frac{n}{2} \log \left(1 + \frac{1}{n} \sum_i \frac{P_i}{\sigma^2} \right) + n\epsilon_n \\
 &\leq \frac{n}{2} \log \left(1 + \frac{P}{\sigma^2} \right) + n\epsilon_n
 \end{aligned}$$

Since $\epsilon_n \rightarrow 0$, we have

$$R \leq \frac{1}{2} \log \left(1 + \frac{P}{\sigma^2} \right).$$